

TCP/IP Communication in Action

Tracing what happens when a user opens a secure website (<https://x.com>)

TEAM-08 | WADF105 | FCDF-COHORT-11 | Level 1, Semester 1
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Introduction and scenario

Adeoti Suliat Adedamola

Scenario: opening <https://x.com>, traced from the sending device to the destination and back.

We cover:

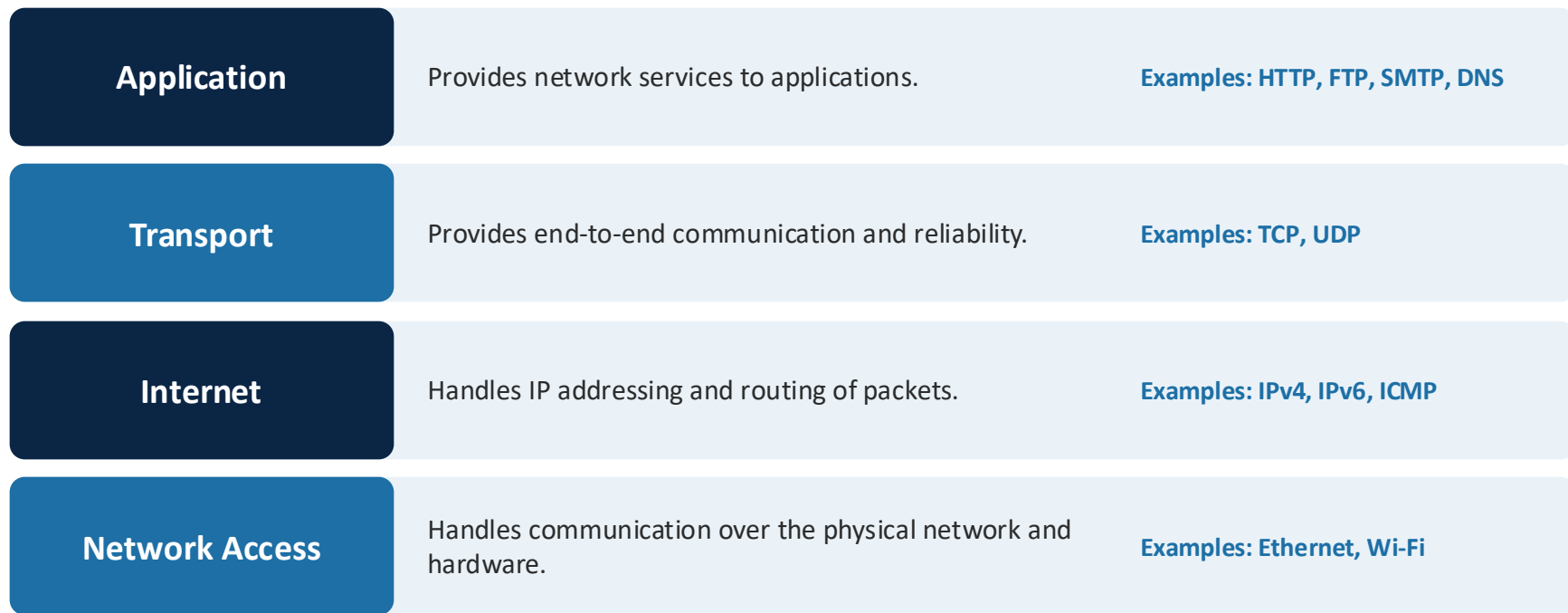
- The four TCP/IP layers and TCP/IP to OSI mapping
- Encapsulation, decapsulation and network devices
- IP addresses, MAC addresses, port numbers and protocols
- TCP vs UDP
- One communication failure and its troubleshooting

Evidence comes from real Wireshark captures on our own lab machines. Lab networks differ, so addresses differ, but the protocol behaviour is the same. Each slide names its capture.

Member	Section
Adeoti Suliat Adedamola (Team Lead)	Introduction, conclusion, coordination
Eyinade Adegoke	TCP/IP layers, OSI, mapping
Kenneth Ukata	Packet journey, devices, encapsulation and decapsulation
Mustapha Abubakar	Protocols, IP/MAC addressing, ports, analysis table
Samina Parveen	TCP vs UDP, failure, troubleshooting

The TCP/IP model: four layers

Eyinade Adegoke



Easy way to describe it: Application → Transport → Internet → Network Access

OSI model and TCP/IP to OSI mapping

Eyinade Adegoke

The OSI (Open Systems Interconnection) Model is a 7-layer framework that standardizes how data moves between computers over a network.

#	Layer	Simple explanation	Examples
7	Application	Provides network services directly to users/applications	HTTP, FTP, SMTP, DNS
6	Presentation	Translates, encrypts, decrypts, and compresses data	JPEG, SSL/TLS, ASCII
5	Session	Establishes, manages, and terminates communication sessions	Session management
4	Transport	Provides reliable data delivery and controls data flow	TCP, UDP
3	Network	Handles logical addressing and routing	IP, routers
2	Data Link	Handles frames, MAC addresses, and local network delivery	Ethernet, switches
1	Physical	Transmits raw bits through cables, radio, or other physical media	Cables, fiber, radio

Easy mapping

TCP/IP Application

→ OSI Application + Presentation + Session

TCP/IP Transport

→ OSI Transport

TCP/IP Internet

→ OSI Network

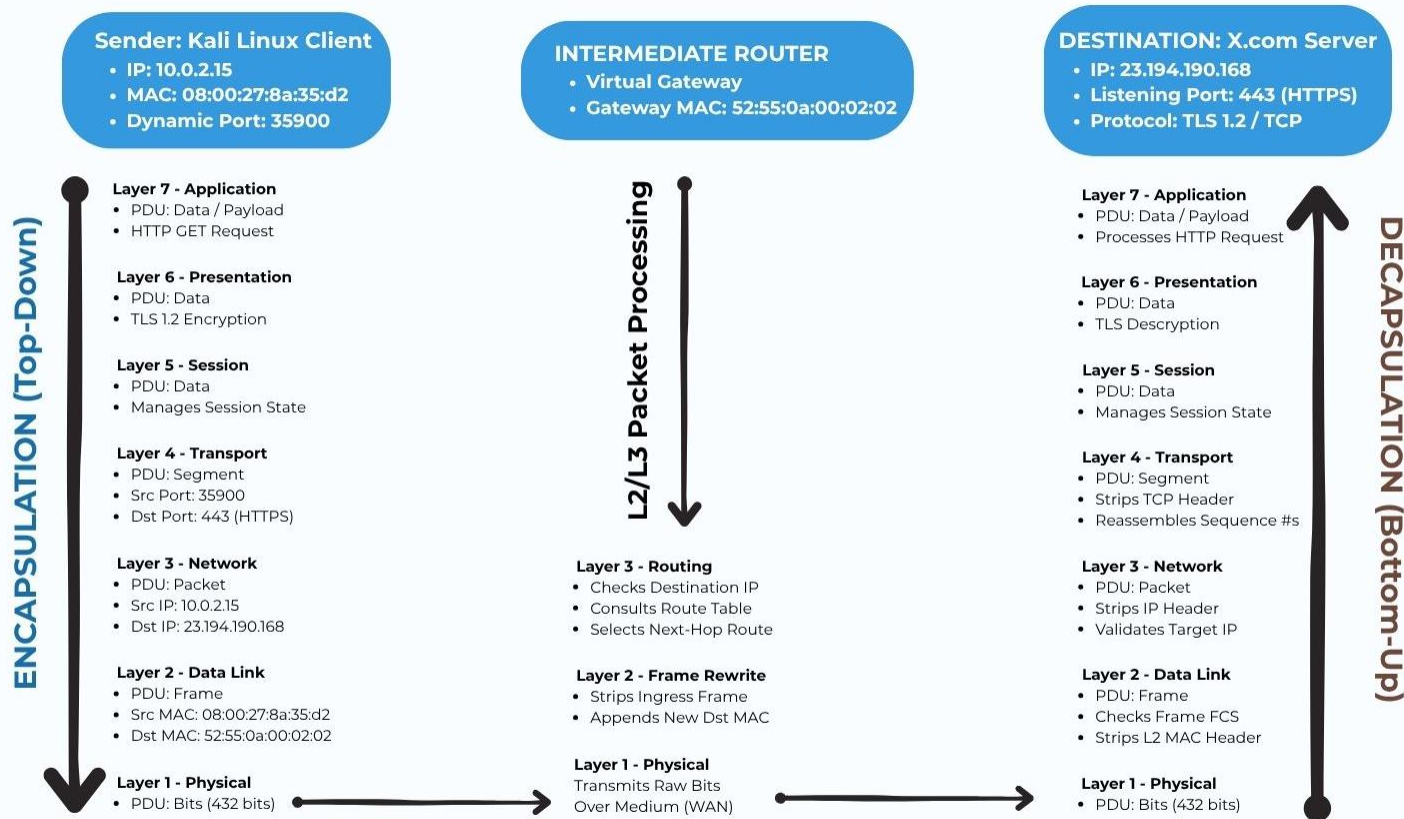
TCP/IP Network Access

→ OSI Data Link + Physical

Key point: The TCP/IP model combines some of the OSI layers, making it a simpler practical model for understanding Internet communication.

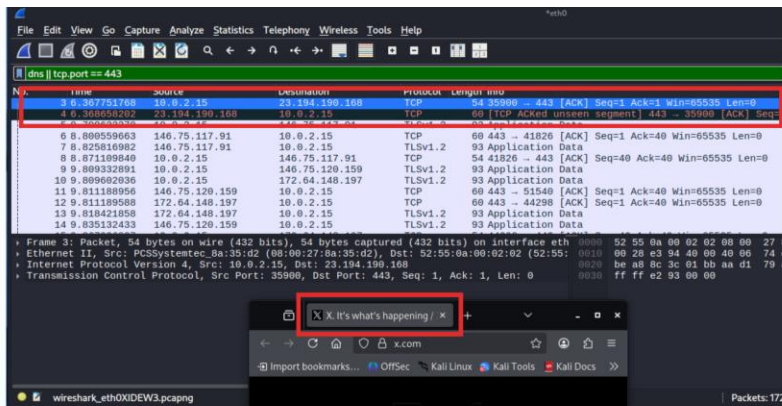
Packet journey: encapsulation and decapsulation

Kenneth Ukata



Network devices and Wireshark evidence

Kenneth Ukata



Role of Network Devices:

Role of network devices

Switch	Router
Layer 2	Layer 3
Forward frames within local segments using MAC tables without altering headers.	Decapsulate incoming Layer 2 Ethernet frames, inspect Layer 3 IP headers to route packets across WANs, and re-encapsulate them with new MAC headers.

OSI encapsulation stack (Packet #3 data)

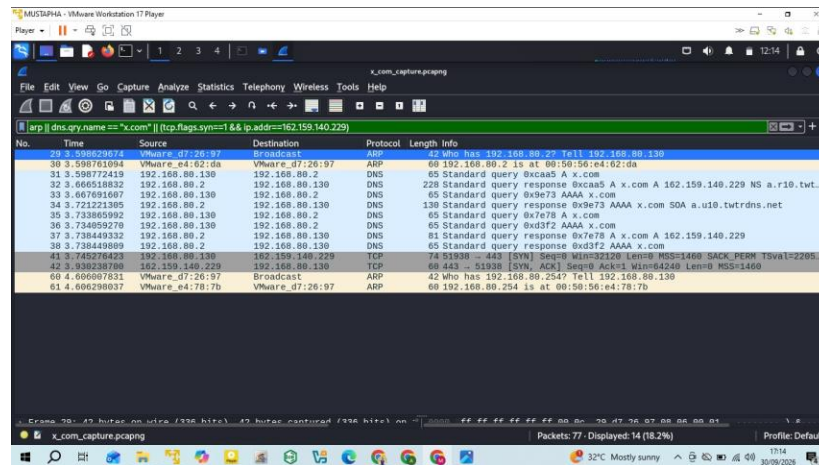
- Layers 7–5: Encrypted TLS 1.2 Web Data.
- Layer 4 (Transport): TCP Header → Src Port 35900, Dst Port 443.
- Layer 3 (Network): IP Header → Src IP 10.0.2.15, Dst IP 23.194.190.168.
- Layer 2 (Data Link): Ethernet II Header → Src MAC 08:00:27:8a:35:d2, Gateway Dst MAC 52:55:0a:00:02:02.
- Layer 1 (Physical): Frame 3 → 54 bytes on wire (432 bits).

Protocols and addressing: opening x.com

Mustapha Abubakar

Protocol order

1. ARP: asks "who has 192.168.80.2?" and the gateway replies with its MAC.
2. DNS (UDP 53): x.com resolved to 162.159.140.229.
3. TCP (port 443): three-way handshake SYN, SYN-ACK, ACK.
4. TLS 1.3 and HTTP/2: connection encrypted, then the request is answered with 200.



No.	Time	Source	Destination	Protocol	Length	Info
29	1.598629674	VMware_e4:62:da	Broadcast	ARP	62	Who has 192.168.80.2? Tell 192.168.80.130
30	3.598761094	VMware_e4:62:da	VMware_d7:26:97	ARP	60	192.168.80.2 is at 00:50:56:e4:62:da
31	3.598772419	192.168.80.130	192.168.80.2	DNS	65	Standard query 0xc0a5 A x.com
32	3.666518832	192.168.80.2	192.168.80.130	DNS	228	Standard query response 0xc0a5 A x.com A 162.159.140.229 NS a.r10.twt...
33	3.667691607	192.168.80.130	192.168.80.2	DNS	65	Standard query 0x9e73 AAAA x.com
34	3.722221305	192.168.80.2	192.168.80.130	DNS	130	Standard query response 0xb073 AAAA x.com SOA a.u10.twtdns.net
35	3.733865992	192.168.80.130	192.168.80.2	DNS	65	Standard query 0x7e78 A x.com
36	3.734859270	192.168.80.130	192.168.80.2	DNS	65	Standard query 0xd3f2 AAAA x.com
37	3.738449332	192.168.80.2	192.168.80.130	DNS	81	Standard query response 0x7e78 A x.com A 162.159.140.229
38	3.738449809	192.168.80.2	192.168.80.130	DNS	65	Standard query response 0xd3f2 AAAA x.com
41	3.748276423	192.168.80.130	162.159.140.229	TCP	74	51938 → 443 [SYN] Seq=161120 Len=0 MSS=1460 SACK_PERM=1 TSval=2290...
42	3.808238708	162.159.140.229	192.168.80.130	TCP	60	443 → 51938 [SYN, ACK] Seq=161120 Len=0 MSS=1460 SACK_PERM=1 TSval=2290...
60	4.606607831	VMware_d7:26:97	Broadcast	ARP	42	Who has 192.168.80.254? Tell 192.168.80.130
61	4.606298037	VMware_e4:78:7b	VMware_d7:26:97	ARP	60	192.168.80.254 is at 00:50:56:e4:78:7b

MAC address

Local link, changes at each hop

00:0c:29:d7:26:97 → 00:50:56:e4:62:da (Kali
→ gateway)

IP address

End to end, stays the same

192.168.80.130 → 162.159.140.229

Port number

Identifies the application

Temporary source port 51938 → 443 (HTTPS).
DNS: 35292 → 53

Protocol analysis table

Mustapha Abubakar

TCP/IP layer	Protocol	PDU	Addressing information	Purpose
Application	DNS	Message	UDP 35292 → 53	Resolves x.com to an IP address
Application	HTTP/2 over TLS 1.3	Message	TCP 51938 → 443	Encrypted web request and response
Transport	UDP	Datagram	Source and destination ports	Fast, connectionless delivery (DNS)
Transport	TCP	Segment	Ports, sequence and ACK numbers	Reliable, ordered delivery (HTTPS)
Internet	IPv4	Packet	192.168.80.130 → 162.159.140.229	End-to-end addressing and routing
Internet	ICMP	Packet	192.168.80.130 → 192.168.80.2	Connectivity test (ping to gateway)
Network Access	Ethernet II	Frame	00:0c:29:d7:26:97 → 00:50:56:e4:62:da	Delivery on the local link
Network Access	ARP	Frame	Broadcast: who has 192.168.80.2?	Finds the gateway MAC address

TCP vs UDP

Two transport protocols, both seen in one secure website visit

Feature	TCP	UDP
Connection	Handshake first	None, sends at once
Reliability	ACKs + retransmission	No guarantee
Ordering	Sequence numbers	No ordering
Header / PDU	20+ bytes / segment	8 bytes / datagram
Speed	Slower, more overhead	Fast, low overhead
In my capture	HTTPS, port 443	DNS, port 53

```
> Frame 11: 65 bytes on wire (520 bits), 65 bytes captured (520 bits) on interface \Device\NPF_{65889820-B41C-4308-FF3665A82D5A}, id 0
> Ethernet II, Src: Intel_41:7e:95 (2c:db:07:41:7e:95), Dst: FiberhomeTel_94:70:a0 (ec:e6:a2:94:70:a0)
> Internet Protocol Version 4, Src: 192.168.1.25, Dst: 192.168.1.1
  > User Datagram Protocol, Src Port: 57452, Dst Port: 53
    > Source Port: 57452
    > Destination Port: 53
    > Length: 31
    > Checksum: 0x839b [unverified]
    > [Checksum Status: Unverified]
    > [Stream Index: 3]
    > [Stream Packet Number: 1]
  > [Timestamps]
    > [Time since first frame: 0.000000000 seconds]
    > [Time since previous frame: 0.000000000 seconds]
  > UDP payload (23 bytes)
  > Domain Name System (Query)
    > Transaction ID: 0x0002
    > Flags: 0x0100 Standard query
    > Questions: 1
    > Answer RRs: 0
    > Authority RRs: 0
    > Additional RRs: 0
    > Queries
      > [Response In: 12]
```

UDP: DNS query, source port 57452 to port 53, length 31

58.3%

packets were TCP (HTTPS/TLS)

3.1%

packets were UDP (DNS)

TCP session to 172.66.0.227:443: SYN, SYN-ACK, ACK, then TLS
Client Hello (SNI = x.com).

Application examples

Samina Parveen

The right protocol depends on what the application can tolerate

TCP · Reliability first

Secure web (HTTPS, 443)

every byte must arrive in order

Email (SMTP, IMAP)

messages must not be lost

File transfer (FTP, SFTP)

one missing block corrupts the file

Remote access (SSH, RDP)

commands must be exact

UDP · Speed first

DNS lookups (53)

one small question, one small answer

Video meetings, VoIP

a late packet is useless

Live streaming, gaming

current data beats retransmission

DHCP, NTP

short request and reply

Communication failure: DNS not responding

Samina Parveen

Website cannot be opened, but the network is healthy

Symptom: "Could not resolve host: x.com"

Cause: DNS server set to 192.0.2.1, where nothing answers

Layer: Application (DNS over UDP port 53)

Key clue: ping 8.8.8.8 still works, so IP and routing are fine

Wireshark: repeated DNS queries to 192.0.2.1, no reply

```
C:\Windows\System32>ping x.com -n 2
Ping request could not find host x.com. Please check the name and try again.

C:\Windows\System32>curl.exe -I https://x.com
curl: (6) Could not resolve host: x.com
```

ip.dst == 192.0.2.1						
No.	Time	Source	Destination	Protocol	Length	Info
19	25.790760	192.168.1.25	192.0.2.1	DNS	65	Standard query 0xb2d1 HTTPS x.com
20	25.791463	192.168.1.25	192.0.2.1	DNS	65	Standard query 0xf9c3 AAAA x.com
21	25.792185	192.168.1.25	192.0.2.1	DNS	65	Standard query 0x0db4 A x.com
22	26.810664	192.168.1.25	192.0.2.1	DNS	65	Standard query 0xc2a2 HTTPS x.com
23	26.810992	192.168.1.25	192.0.2.1	DNS	65	Standard query 0x0454 AAAA x.com
24	26.811282	192.168.1.25	192.0.2.1	DNS	65	Standard query 0xa679 A x.com
25	27.910612	192.168.1.25	192.0.2.1	DNS	76	Standard query 0xd8f4 HTTPS gator.volces.com
26	27.910785	192.168.1.25	192.0.2.1	DNS	76	Standard query 0x2629 AAAA gator.volces.com
27	27.910917	192.168.1.25	192.0.2.1	DNS	76	Standard query 0x239d A gator.volces.com

Second test, port 444 blocked: SYN sent and retransmitted 4 times, never answered. A firewall is likely dropping it.

Troubleshooting procedure

Samina Parveen

Bottom-up: test one layer at a time, then fix and verify

1

Record the symptom

Website fails with "Could not resolve host"

2

ping 8.8.8.8

Replies: internet and routing work

3

nslookup x.com

Fails: name resolution is the fault

4

ipconfig /all

DNS server is 192.0.2.1, not the router

5

Wireshark capture

DNS queries sent, no replies

6

Fix and verify

Reset DNS to 192.168.1.1, flush cache, retest

Verified: gateway ping, 8.8.8.8 ping, nslookup, port 443 test, tracert (8 hops), curl HTTP 200 OK

Conclusion

- Opening x.com is a chain: ARP → DNS (UDP 53) → TCP handshake → TLS → HTTP, each step depending on the one before it.
- Each layer adds its own header: ports (TCP/UDP), IP addresses (end to end), MAC addresses (hop by hop).
- TCP gives reliability; UDP gives speed. One website visit uses both.
- A DNS fault can look like a total outage while IP and TCP are healthy. Test layer by layer, bottom-up.
- Every claim in this presentation is backed by packet evidence.

Thank you. We are happy to take questions.

Team contribution record

TEAM-08

Member	Role	Completed work
Adeoti Suliat Adedamola	Team Lead	Introduction, conclusion, review, submission
Eyinade Adegoke	TCP/IP and OSI	TCP/IP layers, OSI model, mapping
Kenneth Ukata	Packet journey	Packet journey diagram, devices, encapsulation and decapsulation
Mustapha Abubakar	Protocols and addressing	Protocols, IP/MAC, ports, analysis table, 15-figure report, Wireshark capture, deck compilation
Samina Parveen	TCP vs UDP, troubleshooting	TCP vs UDP, examples, DNS failure, troubleshooting steps

Each member's individual evidence link is recorded in their own contribution declaration on the portal.

AI use declaration: Artificial intelligence tools were used by team members for drafting, design and compilation assistance. The team reviewed all content for accuracy and remains responsible for it.

References

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